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Monsoon, 2025

B. Tech 3rd Semester Examination

Civil Engineering, Materials, Testing & Evaluation

Course Code: BCE23302T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

SECTION 1

Answer all the questions **Marks: 10 x 1 =10**

1. (a) What is the main purpose of adding gypsum to cement?
   i. To increase strength
   ii. To reduce setting time
   iii. To prevent cracking
   iv. To control the setting time

   (b) Which test is performed to determine the compressive strength of cement?
   i. Fineness test
   ii. Soundness test
   iii. Consistency test
   iv. Crushing test

   (c) What is the primary function of aggregates in concrete?
   i. To improve the thermal insulation properties
   ii. To reduce the cost of concrete
   iii. To enhance the strength and durability of concrete
   iv. To prevent the concrete from cracking

   (d) In the context of elasticity, Hooke's Law states that stress is proportional to strain up to the:
   i. Yield point
   ii. Ultimate tensile strength
   iii. Elastic limit
   iv. Fracture point

(e) Which of the following is NOT a characteristic of a material that exhibits elastic behavior?

i. It returns to its original shape after the load is removed.
ii. The material deforms permanently after the load is removed.
iii. It obeys Hooke's Law within its elastic limit.
iv. The material undergoes linear deformation under stress.

(f) Poisson's ratio is defined as the ratio of:

i. Lateral strain to longitudinal strain
ii. Longitudinal strain to lateral strain
iii. Shear stress to shear strain
iv. Longitudinal stress to lateral stress

(g) In a bending test, the material of the specimen is subjected to:

i. Shear stress only
ii. Compressive and tensile stress
iii. Pure bending stress
iv. Axial stress

(h) The ability of a material to resist deformation under a twisting load is known as:

i. Shear modulus
ii. Torsional strength
iii. Bending strength
iv. Compressive strength

(i) The primary purpose of the impact test is to measure a material's:

i. Hardness
ii. Ductility
iii. Toughness
iv. Yield strength

(j) Which of the following materials is used to test the impact value of aggregate?

i. Impact test machine
ii. Ball mill
iii. Vibrating table
iv. Compression testing machine

SECTION 2

2(a) Discuss the various types of cements used in construction and their properties. How do different types of cement affect the strength and durability of concrete? (6+4=10)

OR

2(b) Discuss the various types of concrete used in construction and their properties.
What are the advantages and disadvantages of ceramic? (6+4=10)

SECTION 3

3(a) Explain the stress-strain curve for mild steel and brittle material and describe its different regions. (10)

OR

3(b) Explain the mechanical behaviour of materials under external forces. Discuss in detail the key mechanical characteristics such as elasticity, plasticity, ductility, toughness, and hardness, and describe how these properties influence material selection in engineering applications. (10)

SECTION 4

4 (a) Explain the torsion tests used to determine the material strength of a specimen and describe its procedure, the expected behavior of a material under these loads, and the key factors influencing the material's strength. (10)

OR

4 (b) Define creep in materials and explain the stages of creep deformation (primary, secondary, and tertiary). What are the factors affecting creep. (10)

SECTION 5

5 (a) Explain the concept of elastic deformation in materials. How does it differ from plastic deformation? (10)

OR

5 (b) Describe the various types of mechanical tests commonly conducted in a material testing laboratory and explain the purpose of each test. (10)